



PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 3

Introduction to OOPS Programming

Mini Project 2 :

“Creator Dashboard Lite” Objective: Create a console app for digital creators to manage content ideas.

Trainer Guidelines: Implement Content class (title, platform, views, status). Allow user to add, edit, delete content items. Store data using file handling.

TASKS :

1. Create a Content class with properties: title, platform, views, and status. Write a method to display all details of a Content object.



```
#include <iostream>
#include <string>
using namespace std;

class Content {
public:
    string title;
    string platform;
    int views;
    string status;

    // Method to display content details
    void displayContent() {
        cout << "Title: " << title << endl;
        cout << "Platform: " << platform << endl;
        cout << "Views: " << views << endl;
        cout << "Status: " << status << endl;
    }
};

int main() {

    Content content;
```

```

    content.title = "C++ Tutorial";
    content.platform = "YouTube";
    content.views = 1000;
    content.status = "Published";

    content.displayContent();

    return 0;
}

```

OUTPUT:

```

Title: C++ Tutorial
Platform: YouTube
Views: 1000
Status: Published

```

2. Build a console menu that lets users add new content ideas (title, platform, views, status) and save each entry to a text file named content_list.txt.

Hint: Use file handling to append each new content item to the file.



```

#include <iostream>
#include <fstream>
#include <string>
using namespace std;

class Content {
public:
    string title;
    string platform;
    int views;
    string status;
};

int main() {

    int choice;

    cout << "----- Content Manager -----" << endl;
    cout << "1. Add Content" << endl;
    cout << "2. Exit" << endl;
    cout << "Enter your choice: ";
    cin >> choice;
    cin.ignore();

    if (choice == 1) {

```

```

    Content content;

    cout << "Enter title: ";
    getline(cin, content.title);

    cout << "Enter platform: ";
    getline(cin, content.platform);

    cout << "Enter views: ";
    cin >> content.views;
    cin.ignore();

    cout << "Enter status: ";
    getline(cin, content.status);

    // Open file in append mode
    ofstream file("content_list.txt", ios::app);

    file << content.title << endl;
    file << content.platform << endl;
    file << content.views << endl;
    file << content.status << endl;

    file.close();

    cout << "Content saved successfully." << endl;
}

else if (choice == 2) {
    cout << "Program ended." << endl;
}

else {
    cout << "Invalid choice." << endl;
}

return 0;
}

```

OUTPUT:

```

----- Content Manager -----
1. Add Content
2. Exit
Enter your choice: 1
Enter title: assignment
Enter platform: twitter
Enter views: 300
Enter status: uploaded
Content saved successfully.

```

3. Implement a function to read all content items from content_list.txt and display them in a numbered list, showing title and platform for each.



```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

class Content {
public:
    string title;
    string platform;
    int views;
    string status;
};

int main() {

    ifstream file("content_list.txt");

    Content content;
    int number = 1;

    cout << "----- Content List -----" << endl;

    while (getline(file, content.title)) {

        getline(file, content.platform);

        file >> content.views;
        file.ignore();

        getline(file, content.status);

        cout << number << ". "
             << content.title
             << " - "
             << content.platform
             << endl;

        number++;
    }

    file.close();

    return 0;
}
```

OUTPUT:

```
----- Content List -----  
1. assignment - twitter  
2. Minecraft - youtube  
3. life_style - instagram
```

4. Add an option in your console app to let users update the status of any content idea by selecting its number from the displayed list and saving the change back to the file.

Hint: You will need to read all items, modify the selected one, and overwrite the file.



```
#include <iostream>  
#include <fstream>  
#include <string>  
using namespace std;  
  
class Content {  
public:  
    string title;  
    string platform;  
    int views;  
    string status;  
};  
  
int main() {  
  
    Content contents[100];  
    int count = 0;  
  
    // Read all content from file  
    ifstream file("content_list.txt");  
  
    while (getline(file, contents[count].title)) {  
  
        getline(file, contents[count].platform);  
  
        file >> contents[count].views;  
        file.ignore();  
  
        getline(file, contents[count].status);  
  
        count++;  
    }  
  
    file.close();  
  
    // Display content list
```

```

cout << "----- Content List -----" << endl;

for (int i = 0; i < count; i++) {
    cout << i + 1 << ". "
        << contents[i].title
        << " - "
        << contents[i].platform
        << " - "
        << contents[i].status
        << endl;
}

// Select content
int choice;

cout << endl;
cout << "Enter content number to update: ";
cin >> choice;
cin.ignore();

if (choice < 1 || choice > count) {
    cout << "Invalid content number." << endl;
    return 0;
}

// Enter new status
string newStatus;

cout << "Enter new status: ";
getline(cin, newStatus);

contents[choice - 1].status = newStatus;

// Overwrite the file
ofstream outFile("content_list.txt");

for (int i = 0; i < count; i++) {

    outFile << contents[i].title << endl;
    outFile << contents[i].platform << endl;
    outFile << contents[i].views << endl;
    outFile << contents[i].status << endl;
}

outFile.close();

cout << "Status updated successfully." << endl;

return 0;
}

```

OUTPUT:

```
----- Content List -----
1. assignment - twitter - uploaded
2. Minecraft - youtube - uploaded
3. life_style - instagram - production

Enter content number to update: 3
Enter new status: uploaded
Status updated successfully.
```

5. Add a feature to delete a content item by its number from the list, update the file accordingly, and display the updated list to confirm deletion.



```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

class Content {
public:
    string title;
    string platform;
    int views;
    string status;
};

int main() {

    Content contents[100];
    int count = 0;

    // Read all content
    ifstream file("content_list.txt");

    while (getline(file, contents[count].title)) {

        getline(file, contents[count].platform);

        file >> contents[count].views;
        file.ignore();

        getline(file, contents[count].status);

        count++;
    }

    file.close();
```

```

// Display content
cout << "----- Content List -----" << endl;

for (int i = 0; i < count; i++) {

    cout << i + 1 << ". "
        << contents[i].title
        << " - "
        << contents[i].platform
        << endl;
}

// Select content to delete
int choice;

cout << endl;
cout << "Enter content number to delete: ";
cin >> choice;

if (choice < 1 || choice > count) {
    cout << "Invalid content number." << endl;
    return 0;
}

// Shift remaining elements
for (int i = choice - 1; i < count - 1; i++) {
    contents[i] = contents[i + 1];
}

count--;

// Overwrite the file
ofstream outFile("content_list.txt");

for (int i = 0; i < count; i++) {

    outFile << contents[i].title << endl;
    outFile << contents[i].platform << endl;
    outFile << contents[i].views << endl;
    outFile << contents[i].status << endl;
}

outFile.close();

cout << endl;
cout << "Content deleted successfully." << endl;

// Display updated list
cout << endl;
cout << "----- Updated Content List -----" << endl;

```



```
for (int i = 0; i < count; i++) {  
  
    cout << i + 1 << ". "  
        << contents[i].title  
        << " - "  
        << contents[i].platform  
        << endl;  
}  
  
return 0;  
}
```

OUTPUT:

```
----- Content List -----  
1. assignment - twitter  
2. Minecraft - youtube  
3. life_style - instagram  
  
Enter content number to delete: 3  
  
Content deleted successfully.  
  
----- Updated Content List -----  
1. assignment - twitter  
2. Minecraft - youtube
```